

OFFICIAL TECHNICAL DOCUMENTATION

MCA WALLAH

Full-Stack Portfolio & CMS Engine

A comprehensive architectural blueprint, component design, database model specifications, REST API reference, and administrative maintenance manual.

AUTHOR / CREATOR

Mahadeb Maity

PROJECT VERSION

1.0.0 (Production Release)

TECHNOLOGY STACK

**React 19, Vite, Node.js, Express,
MongoDB**

TARGET AUDIENCE

**Software Engineers, Tech Recruiters &
Clients**

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1. Executive Summary & Purpose

1.1 Project Vision

The **MCA WALLAH Portfolio** is an enterprise-grade personal branding platform engineered to demonstrate high-level full-stack development expertise. Unlike static portfolio templates, this application is a dynamic, living web system powered by an autonomous RESTful backend, persistent MongoDB database, and a full-featured administrative studio (CMS).

1.2 Core Business & Technical Objectives

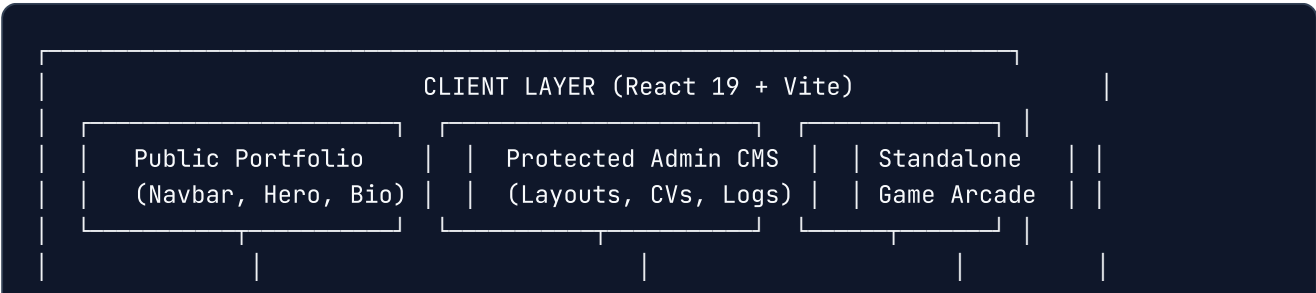
- **Dynamic Content Customization:** Eliminate hardcoded content by enabling real-time administrative modifications for every visual section (Hero, Navbar, Resume Vault, Skills, Projects, and Games).
- **Modern Visual Excellence:** Deliver state-of-the-art UI/UX using custom CSS design tokens, frosted glassmorphism (`backdrop-filter: blur(28px)`), and dynamic responsive layout switching.
- **Interactive Engagement:** Maximize visitor retention through an interactive Retro Gaming Arcade (Snake, Unbeatable Minimax AI Tic-Tac-Toe, Typing Challenge, 15-Puzzle) and a movable Sticky Background Engine.
- **Permanent Resume Library:** Maintain a secure repository of candidate CVs and resumes with dynamic button toggling and multi-resume inline preview/download capabilities.

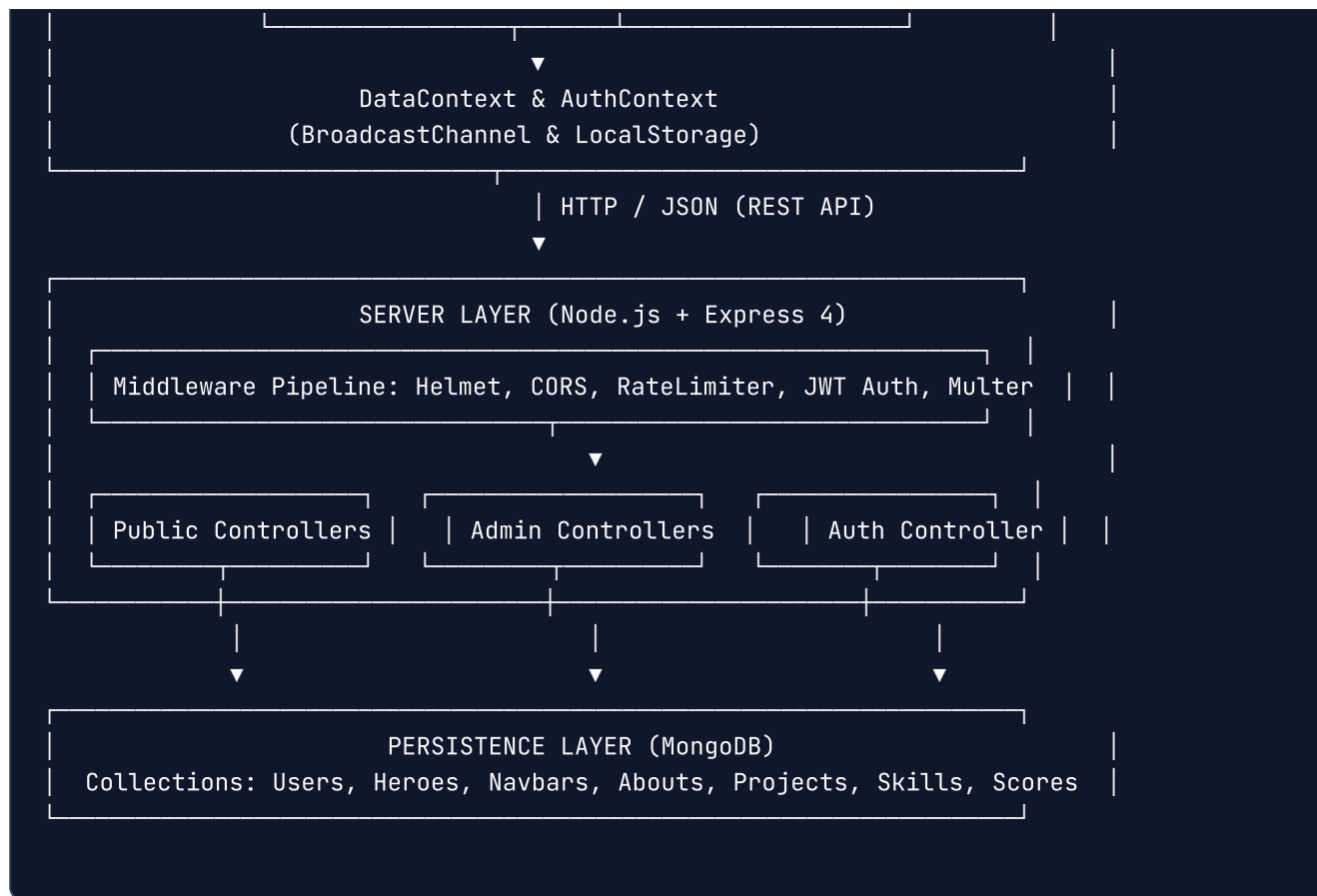
Key Innovation: Real-Time Multi-Tab Synchronization

The application employs the modern `BroadcastChannel` API alongside optimistic React state updates. Changes made in the Admin CMS reflect instantly on open portfolio visitor tabs without requiring full page reloads.

2. System Architecture & High-Level Design

2.1 High-Level Architecture Diagram





3. Frontend Architecture & Design Systems

3.1 Dynamic Navigation Engine

The navigation bar is built with an adaptive auto-fit geometry (`width: max-content; max-width: min(96vw, 1320px)`) that prevents element clipping and supports 4 distinct presentation presets configurable from the Admin panel:

Preset ID	Design Aesthetic	Visual Attributes
<code>floating-dock</code>	macOS Frosted Glass Island	Central capsule, <code>backdrop-filter: blur(28px)</code> , subtle shadow aura
<code>cyber-capsule</code>	Cyberpunk Neo-HUD	High-contrast dark base, glowing neon border, cyan/magenta accents
<code>minimal-island</code>	Ultra-Clean Compact Pill	Streamlined geometry, auto-expanding pill boundary, tight padding
<code>full-width</code>	Classic Enterprise Sticky Bar	Edge-to-edge full width with bottom border and glass background

3.2 Multi-Style Hero Section

The Hero section accommodates diverse candidate identities through 4 structural layout engines:

-  **Glassmorphic Floating Capsule:** Centered glassmorphic card with dynamic stat chips and call-to-action buttons.
-  **Two-Column Developer Split:** Left-hand copy paired with a live interactive macOS Code Terminal with syntax highlighting and 1-click clipboard copying.
-  **Minimalist Editorial Spotlight:** Bold high-impact typography with subtle radial ambient backdrops.
-  **Retro-Futuristic Hologram:** Angular cyberpunk badges, technical coordinates, and animated scanlines.

3.3 Draggable Floating Backdrop Widget (Background)

Visitors can freely reposition the theme customization widget anywhere on the desktop or mobile viewport via dual mouse/touch event listeners. Coordinates are saved in `localStorage` so the preferred widget position persists across browser visits.

4. Permanent Multi-Resume Vault Engine

The resume subsystem solves the limitation of single-file static PDF links:

- **Permanent Cloud Vault:** Every resume uploaded via Admin Studio is permanently archived in the backend storage with filename, file size, and upload timestamp.
- **Multi-Button Visibility Controller:** The administrator can dynamically toggle visibility for up to **3 concurrent resume buttons** (e.g. *Full Stack CV*, *Frontend Resume*, *Academic Bio*).
- **Glassmorphic Dropdown & Inline Preview:** If multiple resumes are active, the header seamlessly transforms into an interactive dropdown with direct download () and new-tab preview () capabilities.

5. Interactive Gaming Arcade & Minimax AI

5.1 Arcade Games Overview

Game Title	Type & Mechanism	Algorithmic / Technical Highlight
 Retro Snake	Grid-based Canvas Game	Tick-rate velocity scaling, collision detection, and score multiplier.
 Tic-Tac-Toe AI	Strategy Board Game	Unbeatable Minimax Algorithm with recursive alpha-beta tree evaluation.
 Speed Typing	Developer WPM Challenge	Character-by-character accuracy parser, real-time WPM calculation, timer hook.
 15-Puzzle Sliding	Permutation Grid Puzzle	Solvability inversion checker, swap matrix, move counter.

5.2 Dual Window Mode (Inline & Standalone Window)

Games can be played inside the interactive modal or launched into an isolated, distraction-free standalone window at `/arcade`. High scores are automatically synced to the MongoDB global leaderboard.

6. Protected Admin Studio CMS Architecture

6.1 Administrative Modules Overview

Panel Route	Module Name	Administrative Capabilities
<code>/admin/dashboard</code>	 Analytics Overview	Real-time visitor counts, message analytics, and system status indicators.
<code>/admin/navbar</code>	 Navbar Studio	Switch layout presets, customize brand prefix/suffix, style "Let's Talk" CTA.
<code>/admin/hero</code>	 Hero & Header CMS	Change hero layout modes, edit typewriter roles, modify CTA anchors and pills.
<code>/admin/about</code>	 Profile & Resume Vault	Upload permanent resumes, manage button visibility limits, edit bio and stats.

Panel Route	Module Name	Administrative Capabilities
/admin/projects	 Projects Manager	Create/Edit projects, assign tech tags, upload preview cards, manage order.
/admin/skills	 Skills Matrix	Add skill icons, assign categories (Frontend/Backend/DevOps), set proficiency %.
/admin/games	 Arcade & Leaderboard	Manage active games, descriptions, and moderate leaderboard high scores.
/admin/messages	 Feedback Inbox	Read, mark as read, and delete contact form inquiries.
/admin/settings	 Account & Credentials	Change Admin username, email, password, and export JSON backups.

7. Database Schemas & Data Modeling

7.1 Core Mongoose Schemas Summary

```
// 1. User & Admin Schema
const UserSchema = new Schema({
  name: { type: String, required: true },
  email: { type: String, required: true, unique: true, lowercase: true },
  password: { type: String, required: true }, // BCrypt Hashed
  role: { type: String, enum: ['user', 'admin'], default: 'user' },
  avatar: { type: String },
  preferences: {
    darkMode: { type: Boolean, default: true },
    background: { type: String, default: 'mesh' },
    accentColor: { type: String, default: '#e84545' }
  }
}, { timestamps: true });

// 2. Navbar Schema
const NavbarSchema = new Schema({
  layoutStyle: { type: String, default: 'floating-dock' },
  logoText: { type: String, default: 'Mahadeb' },
  logoPrefix: { type: String, default: '<' },
  logoSuffix: { type: String, default: '>' },
  showHireMeButton: { type: Boolean, default: true },
  hireMeButtonText: { type: String, default: "Let's Talk" },
  hireMeStyle: { type: String, default: 'gradient-glow' },
  hireMeIcon: { type: String, default: 'fa-solid fa-paper-plane' },
  hireMeTarget: { type: String, default: 'contact' }
}, { timestamps: true });

// 3. About & Resume Vault Schema
const AboutSchema = new Schema({
  bio: [String],
  resumes: [{
    title: String,
    url: String,
    isPrimary: Boolean,
    activeOnPortfolio: Boolean,
    fileSize: String,
    uploadedAt: { type: Date, default: Date.now }
  }]
});
```

8. RESTful API Endpoint Documentation

Method	Endpoint Route	Access Level	Description
GET	/api/portfolio/public	Public	Aggregates all active portfolio data in a single high-performance payload.
POST	/api/portfolio/contact	Public	Validates feedback message and sends automated Nodemailer notification.
POST	/api/auth/login	Public	Authenticates credentials and returns JWT Access & Refresh tokens.
PUT	/api/user/profile	Protected (JWT)	Updates authenticated admin's display name and login email.
PUT	/api/user/password	Protected (JWT)	Verifies current password and updates to new bcrypt-hashed password.
POST	/api/user/reset-defaults	Protected (JWT)	Restores factory default credentials (mahadeb@portfolio.com).
POST	/api/admin/section/:type	Admin Guard	Updates CMS section items in MongoDB and broadcasts cross-tab updates.

9. Security, Authentication & Rate Limiting

9.1 Authentication Architecture

- **JWT Dual-Token Protocol:** Short-lived Access Tokens (15m) paired with rolling Refresh Tokens (7d) stored securely in client storage.
- **BCrypt Hashing:** Passwords salted with 10 rounds before database persistence.
- **Route Guards:** Protected React Router wrapper (`AdminAuthGuard.jsx`) validates role privileges before rendering administrative panels.

9.2 API Protection & Headers

- **Helmet Security Headers:** Configures XSS filters, CSP protections, and prevents iframe clickjacking.
- **Express Rate Limiting:** Prevents DDoS and brute-force attempts on sensitive endpoints like `/api/auth/login` and `/api/portfolio/contact`.

10. Deployment, Production & Troubleshooting Guide

10.1 Vercel SPA Routing Configuration (`vercel.json`)

To prevent `404: NOT_FOUND` errors on client-side routes (`/admin/login` , `/admin/dashboard` , `/arcade`), the root project contains:

```
{
  "rewrites": [
    {
      "source": "/(.*)",
      "destination": "/index.html"
    }
  ]
}
```

10.2 Default Administrative Credentials Quick Reference

Parameter	Default Production Value
Admin Login Portal	<code>http://localhost:5173/admin/login</code>

Parameter	Default Production Value
Default Email	<code>mahadeb@portfolio.com</code>
Default Password	<code>Admin@123456</code>
Reset Mechanism	Admin Studio $\rightarrow$ Settings $\rightarrow$ "Reset to Default Credentials"